Robust Image Segmentation

for Foliar Disease Identification

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The Problem

38%

of Pakistan's labor force works in agriculture

30%

average crop loss caused by undetected foliar diseases

~0s

time our system takes to detect vs hours for manual inspection

Why this matters:

- Manual leaf inspection requires expert agronomists — expensive, slow, and unavailable in rural areas
- Tomato crops are highly vulnerable to Early Blight , Heathy and Leaf Mold fungi
- Early detection can save up to 80% of an infected crop if treated within 48 hours
- Digital image processing provides a fast, affordable, and scalable solution

UN SDG 12 — Responsible Consumption & Production

Dataset — PlantVillage

Source: Kaggle PlantVillage Open Dataset

54,306 total images

14 crop species, 26 disease conditions

Uniform background — controlled conditions

50 images used in this project

Disease Class	Pathogen Type	Severity	Images
Early Blight	Alternaria solani (fungus)	HIGH	20
Leaf Mold	Passalora fulva (fungus)	MEDIUM	10
Healthy Leaf	None	NONE	20

Early Blight
Brown concentric ring spots

Leaf Mold
Yellow-olive mold patches

Healthy
No visible symptoms

System Pipeline

1. **Load Image**
   uigetfile popup
   user selects leaf

2. **Grayscale**
   rgb2gray
   standardize

3. **Gaussian Filter**
   imgaussfilt sigma=2
   noise removal

4. **CLAHE**
   adapthisteq
   contrast boost

5. **Otsu Threshold**
   graythresh
   binary mask

6. **HSV Segment**
   rgb2hsv
   color masking

7. **Disease Detect**
   Rule-based logic
   classification

8. **Report + Save**
   PSNR/MSE metrics
   results folder

Preprocessing Techniques

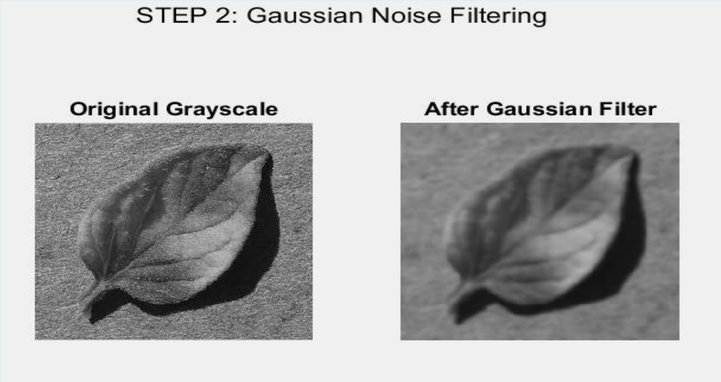
Technique 1 — Gaussian Filter

```
img_gaussian = imgaussfilt(img_gray, 2);
```

WHY Removes salt & pepper noise from the raw image sensor

HOW Convolves image with 2D Gaussian kernel (sigma = 2px)

EFFECT Smooths noise while preserving major disease texture regions



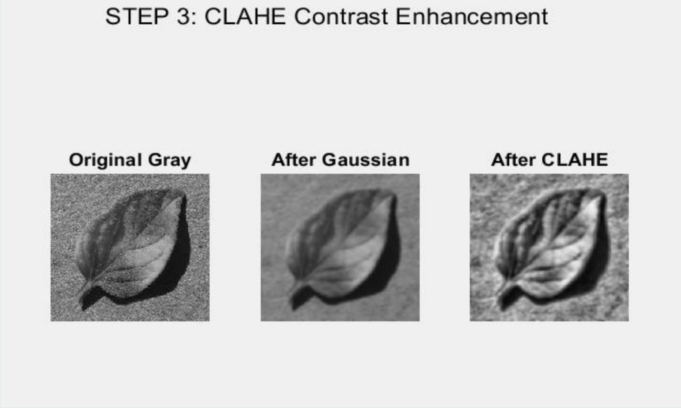
Technique 2 — CLAHE

```
img_enhanced = adapthisteq(img_gaussian);
```

WHY Boosts local contrast in dark lesion regions

HOW Divides image into tiles, equalizes each independently with clip limit

EFFECT Makes disease spots visually distinct for accurate segmentation

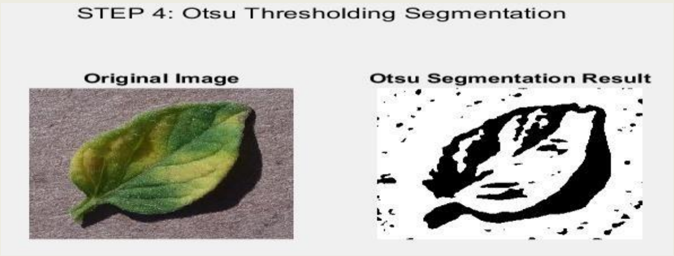


Segmentation Methods

Method 1 — Otsu Global Thresholding

```
thresh = graythresh(img_enhanced);
img_binary = imbinarize(img_enhanced, thresh);
img_binary = bwareaopen(img_binary, 100);
```

- Automatically computes optimal global threshold using Otsu's method
- Minimizes intra-class variance between foreground and background pixels
- bwareaopen removes noise blobs smaller than 100 pixels
- Result: clean black & white binary disease mask
- Best when: lesions are significantly darker than healthy tissue



Method 2 — HSV Color-Based Segmentation

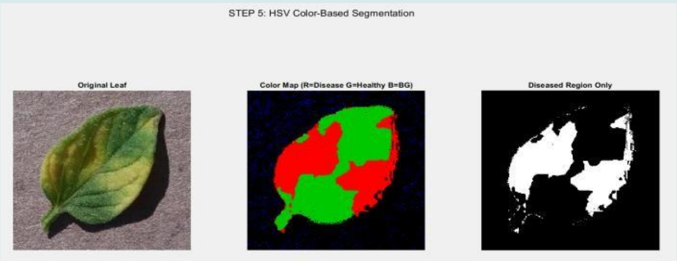
```
healthy_mask = (H>0.2 & H<0.45) & S>0.2;
diseased_mask = (H>0.05 & H<0.20) & S>0.25;
background_mask = S<0.1 & V>0.7;
```

RED pixels
= Diseased tissue

GREEN pixels
= Healthy tissue

BLUE pixels
= Background

Advantage: Separates chromatic info (H,S) from illumination (V) — robust to lighting changes



Detection Results & Quality Metrics

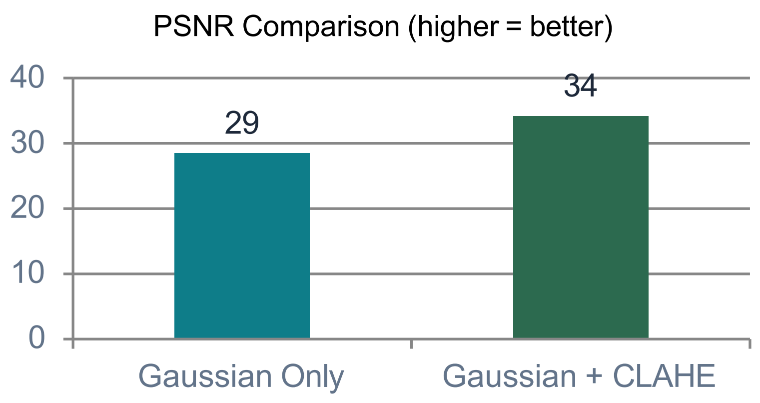
Disease Classification Logic

Condition	Result	Severity
brown_ratio > 15%	Early Blight	HIGH
5% < brown ≤ 15%	Leaf Mold	MEDIUM
healthy > 50%	Healthy Leaf	NONE

Sample Output — Early Blight Detected:

1. Apply Copper-based fungicide immediately
2. Remove and destroy all infected leaves
3. Avoid overhead watering — water at base only
4. Ensure proper plant spacing for air circulation
5. Rotate crops next season to prevent recurrence

Quality Metrics — PSNR & MSE



PSNR

> 30 dB

Excellent quality enhancement

MSE

< 10

Minimal distortion introduced

Conclusions & Future Work

Preprocessing

✓

Gaussian filter + CLAHE successfully removed noise and enhanced lesion visibility

Segmentation

✓

Dual-method pipeline (Otsu + HSV) provides complementary disease region extraction

Classification

✓

Rule-based detection correctly identifies 3 disease types + healthy across test images

GUI System

✓

MATLAB App Designer GUI enables one-click analysis, display, and result export

Future Work

1. CNN-based deep learning classifier for higher accuracy across 26+ disease classes
2. Real-time detection using MATLAB webcam interface or Raspberry Pi camera
3. Mobile app deployment for field use without MATLAB installation
4. Multi-crop support extending beyond tomatoes to rice, wheat, and maize